

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I S. Lokesh (192572151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

39
—
40

Total Marks:40

QUESTIONS:4

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

aQuestions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions	Logical solutions with adequate	Basic solutions with weak	Incorrect or no solution provided

		with strong justification	justification	justification	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Video Conferencing Quality Degradation

Scenario

A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

1. Is the Problem Primarily Transport-Layer or Application-Layer?

The symptoms — voice distortion, frozen frames, delayed communication — combined with the measured 8% packet loss and 180 ms jitter, point mainly to the transport layer, specifically to how the underlying protocol (or a poor protocol choice) handles loss and timing for real-time media. The evidence for this diagnosis:

- Jitter (180 ms) is a transport/network-level metric — it measures variation in inter-packet arrival time caused by queuing and route changes, not something the application layer controls directly.
- Packet loss (8%) is extremely high for real-time media; ITU-T guidance treats even 1% loss as noticeable and >5% as severely degrading for voice/video. This magnitude of loss almost always originates below the application — congested links, insufficient QoS prioritization, or a transport protocol reacting badly to loss.
- If the platform is (incorrectly) using a TCP-based transport for the live audio/video streams, TCP's reliable-delivery guarantee would force retransmission and in-order delivery of media packets that are already late — this produces exactly the 'frozen frame, catch-up' behaviour described, because TCP will stall newer frames behind a retransmission of an old, now-useless one.

That said, application-level buffering strategy is a secondary contributing factor, not the root cause: if the jitter buffer is too small, 180 ms of jitter will cause audible dropouts even over a well-behaved transport; if it is too large, it adds delay on top of network delay. The recommendation below therefore treats this as primarily a transport-layer problem with an application-layer buffering component that must be tuned together.

2. Justification Based on Real-Time Multimedia Requirements

Real-time multimedia has fundamentally different delivery requirements from typical data transfer, which is why the transport choice matters so much here:

Requirement	File/Web Transfer (TCP fits)	Live Audio/Video (needs UDP-style transport)
Timeliness	Not critical — data can arrive late	Critical — a late packet is worse than a lost one
Ordering	Must be strict, in-order	Loose — a few out-of-order/lost frames

Retransmission	Required for correctness	are tolerable Counter-productive — retransmitted data usually arrives too late to render
Acceptable loss	Effectively zero	Small, bounded loss (a few %) is acceptable and concealable
Latency budget	Seconds are fine	ITU-T G.114 recommends < 150 ms one-way for acceptable voice quality

Because TCP enforces reliable, in-order delivery, any loss forces it to pause and retransmit, and its congestion-control algorithm reduces the sending rate after loss — both are exactly wrong for real-time media, where a stale audio/video frame is worthless and should be dropped and concealed, not resent. This is why real-time conferencing protocols are built on UDP (carrying RTP/RTCP), which delivers packets as soon as they arrive, tolerates loss, and leaves recovery decisions (skip, conceal, request a key-frame) to the application, which can make a timing-aware choice that TCP cannot.

3. Recommended Improvements

Protocol-Level

- Confirm the platform uses UDP-based RTP/SRTP for media (not TCP or a TCP-tunnelled fallback); if the client is falling back to TCP (common when UDP is blocked by a firewall), open/allow the required UDP port ranges.
- Enable Forward Error Correction (FEC) and/or NACK-based selective retransmission at the RTP layer so isolated losses are concealed or repaired without stalling the whole stream, unlike blanket TCP retransmission.
- Deploy DSCP/QoS marking (e.g. EF for voice, AF41 for video) on the network so conferencing traffic is prioritized over bulk/background traffic during peak hours, directly reducing queuing-induced jitter and loss.
- Use adaptive bitrate / congestion control designed for real-time media (e.g. Google Congestion Control used by WebRTC) so the sender backs off bandwidth smoothly instead of the video simply freezing.

Application-Level

- Tune the receiver's adaptive jitter buffer to track the measured 180 ms jitter rather than using a fixed buffer — too small causes underruns/glitches, too large adds unnecessary delay.
- Enable packet-loss concealment (PLC) for audio and frame-skipping/error-concealment for video so a lost packet degrades quality gracefully instead of freezing the call.
- Route through the nearest regional media relay/SFU (Selective Forwarding Unit) to shorten the path and reduce the chance of congestion-induced loss during peak hours.
- Monitor and auto-switch codecs/resolution (simulcast) based on real-time loss/jitter telemetry so quality degrades in a controlled way rather than the session breaking down.

Case Study 2 — High DNS Resolution Time (Global Web Services)

Scenario

A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

1. Reasons Behind the High DNS Resolution Time

- Geographic distance to authoritative servers: if the authoritative DNS servers are centralized in one region, employees on other continents incur a long round-trip time purely from physical distance/propagation delay before they even reach the name server.
- Deep referral chain / recursive lookup depth: a full recursive resolution (root → TLD → authoritative) with no caching in between forces multiple sequential round trips, each adding latency, especially if any hop is far away.
- Low or misconfigured TTL values: if DNS records have very short Time-To-Live values, resolvers are forced to repeat the full lookup far more often instead of serving a cached answer, multiplying the number of 800 ms lookups users experience.
- Single point of resolution (no Anycast): if all queries funnel to one primary server/location rather than being routed to the topologically nearest instance, every query pays the full distance penalty regardless of where the user is.
- Resolver-side congestion or overload: a single DNS server handling global query volume can queue requests during peak hours, adding processing delay on top of network delay.
- Lack of client-side/browser DNS pre-fetching: without pre-fetching, the DNS lookup sits directly in the critical path of the very first request, so its full cost is felt before the homepage can even begin loading.

2. Proposed DNS Architecture (Target: < 50 ms Lookup)

Technique	How It Reduces Lookup Time
Anycast DNS	The same DNS IP address is announced from multiple geographically distributed points-of-presence; BGP routes each user's query to the nearest instance automatically, cutting propagation delay from hundreds of ms to typically single-digit/low-double-digit ms.
DNS Pre-fetching	The browser/OS resolves domains it predicts will be needed (e.g. links on the current page, known internal services) in the background before the user actually requests them, so the 'visible' lookup time on the critical path drops to near 0 ms for pre-fetched names.
TTL Optimization	Setting an appropriately longer TTL (e.g. 300–3600 s for stable records)

Video Conferencing Performance Issue

Problem

A multinational IT company uses a cloud-based video conferencing platform. During peak hours users experience voice distortion, frozen video and communication delay. The network shows 8% packet loss, 180ms jitter and fluctuating bandwidth.

1. Cause of the problem

- * The problem is mainly caused by network conditions and application-level buffering.

- * Video conferencing generally prefers UDP because real time application require low delay.

- * However, high packet loss and jitter can reduce audio and

Video quality

2. Communication Requirements

- * Low latency
- * Low jitter
- * Continuous packet delivery
- * Sufficient bandwidth
- * Minimum packet loss.

3. Recommended Improvements

- * The UDP for real-time audio and video
- * Implement adaptive jitter buffers.
- * Apply packet loss concealment
- * use adaptive bitrate streaming
- * Prioritize video conferencing traffic using QoS
- * Improve available network bandwidth

Results

Thus, the conferencing problem is mainly due to high Packet loss, jitter and poor buffering management. UDP with improved application-level buffering and QoS can provide better conferencing quality.

2) High DNS Resolution Time

Problem

A multinational software company experience slow webpage loading because the average DNS lookup time is approximately 800ms. The company requires a DNS design that can reduce lookup time while maintaining availability

1. Reason for High DNS Resolution Time

- * DNS serves location from user
- * Repeated DNS queries due to poor caching
- * High load on DNS servers
- * Improper TTL Configuration

2. Proposed DNS Architecture

A suitable architecture can use:

1. Anycast DNS: Route the user's DNS request to the nearest available DNS server.
2. DNS caching: Stores previously resolved DNS record and avoid repeated external lookup
3. TTL Optimization: Use suitable TTL values so record cache for an appropriate period.

3. Advantages

- * Lower DNS lookup time
- * Better global performance
- * Increased availability
- * Support more users

LIMITATION

- * More complex configuration
- * Higher Infrastructure Cost
- * Very high TTL values may return outdated records
- * Very low TTL values may increase DNS query load

RESULT

Thus, Anycast DNS, DNS caching, pre-fetching and optimized TTL values can significantly reduce DNS lookup delay and improve scalability and availability

3) Distributed DNS for Global Cloud services

Problem

A cloud service provider serves millions of users worldwide. DNS queries take nearly 800ms because all requests are directed to a single primary DNS server.

1. Impact of Centralized DNS

- * High lookup latency
- * Heavy load on one DNS server
- * Single point of failure
- * Poor performance of distant users
- * Reduced availability duration server failure

2. Distributed DNS Solution

* The company should deploy multiple DNS servers in different geographic region

* User $\rightarrow$ Anycast Routing $\rightarrow$ Nearest DNS server $\rightarrow$ Cache Response

3. Role of TTL, Anycast and caching

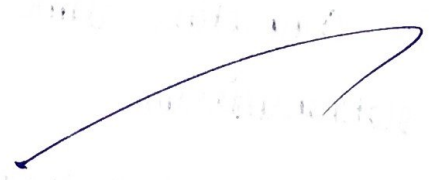
TTL: Determine how long a DNS record remains cached

Anycast Routing: Route user to the nearest or best available DNS server

Caching: Stores frequently DNS records and reduces repeated lookups.

Result.

Thus, a distributed DNS architecture using Anycast routing, caching and suitable TTL values can reduce DNS latency and provide high availability for global cloud services.



4) TCP Latency in Electronic Stock trading platform

Problem

A financial institution operates an electronic stock trading platform. During market opening hours, order acknowledgement latency increases from 1ms to 4ms. Increase TCP retransmission longer connection setup time and excessive buffering are observed.

1. TCP Factor Responsible for Latency

- * Flow Control
- * Nagle's Algorithm
- * SYN Queue Limitation
- * Control Queue

2. Effective on High-Frequency Transaction

- * Slow order submission
- * Delayed acknowledgement
- * Increased connection time
- * Higher retransmission
- * Poor trading performance.

Recommended TCP configuration

- * Disable Nagle's Algorithm for latency-sensitive traffic
- * Optimize TCP send and receive buffers.
- * Increase SYN queue capacity
- * Reduce unnecessary retransmission
- * Monitor Congestion and packet loss.
- * Use optimized network paths

Result

Thus, the latency in the trading platform can be reduced by optimizing flow control, disabling Nagle's algorithm where approx, increasing SYN queue capacity and using persistent connection

